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Frontend Specification Document IPO Allotment Verification System Prepared by: Bhavya Bulani Version 1.2 · July 2026

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Introduction

This document specifies the frontend of the IPO Allotment Verification System, built with React and Tailwind CSS. It covers the screen inventory, key components, states, and interaction flows needed to support the workflows defined in the PRD, including automatic IPO detection, multi-IPO selection, and result caching.

Note on Version 1.2: the backend’s switch from SQLite to MySQL (see the Technical Architecture and Database Design Documents) is an infrastructure-layer change and does not alter any frontend screen, component, or interaction flow described in this document — the frontend talks to the same REST API contract either way. No changes were required in this document as a result of the database migration; the version number is bumped only to keep the document set aligned.

Technology Approach

Aspect	Approach
Framework	React (functional components + hooks)
Styling	Tailwind CSS utility classes
State Management	Local component state plus a lightweight shared store for run status/progress, tracked per-IPO as well as in aggregate
Data Fetching	REST calls to the FastAPI backend; polling (or streaming) for progress updates during a run

Screen Inventory

1. **Mode Selection / Home** — choose Single Client or Bulk (Excel Upload) mode.
2. **Single Client Entry** — enter PAN or Client Code, select one or more IPOs, view result(s).
3. **Bulk Upload** — upload Excel file, select one or more IPOs, review detected columns and validation summary.
4. **Progress Screen** — live progress bar and current-client indicator during a bulk run, with an optional per-IPO breakdown. Includes an inline CAPTCHA Prompt overlay/panel.
5. **Results Screen** — results table with status categories, cache indicators, search, and Excel download.
6. **Dashboard** — summary counts, with a per-IPO breakdown when multiple IPOs were checked.
7. **History** — list of past runs, with the ability to reopen a previous report.
8. **Notifications** — completion toast/banner summarizing a finished run.

Screen Specifications

Mode Selection / Home

- Two clear entry points: “Check Single Client” and “Upload Excel File”.
- The IPO selector supports choosing multiple IPOs.
- Auto-detected IPOs appear automatically, each marked with a small “Auto-detected” badge.

Single Client Entry

Fields — PAN or Client Code input. IPO selector (multi-select). “Check Allotment” action button.

Validation — Inline validation against PAN/Client Code format. At least one IPO must be selected before the button is enabled.

Result Display — When multiple IPOs are selected, the result view shows one row per IPO:

Name	PAN	IPO	Status	Source
Rahul Shah	ABCDE1234F	NSDL IPO	Allotted	Cache
Rahul Shah	ABCDE1234F	HDB Financial IPO	Not Allotted	KFin Technologies

Bulk Upload

Upload control — Accepts .xlsx / .xls only. Enforces the 10,000-row cap.

IPO selection — Multi-select control; selected IPOs shown as removable chips/tags. Inline note clarifies scope (e.g., “1,000 clients × 2 IPOs = up to 2,000 checks”).

Column Detection Summary — Shows detected → mapped column pairs (e.g., “Pan No” → PAN).

Validation Summary — Total rows, valid rows, invalid rows with reason, selected IPO count.

Progress Screen

Checking IPO Allotment



1,340 / 3,000 Checked (2 IPOs selected)

Current Client: Rahul Shah – NSDL IPO
Cache hits so far: 214

- Progress bar reflecting completed/total across all selected IPOs.
- Current client (and IPO, when multiple) being processed.
- Running cache-hit count.
- Optional expandable per-IPO progress breakdown.
- Updates via polling or streaming, not manual refresh.

CAPTCHA Prompt (Assisted Mode) — Shows the CAPTCHA image inline with a text input and Submit button when a registrar presents a challenge for the client currently being checked; other clients/workers continue processing in parallel.

Results Screen

Table Columns: Name, PAN/Client Code, IPO, Status, Source (Registrar or Cache), Time Checked.

Status badges (color-coded): Allotted (green), Not Allotted (red), error states (amber).

A small **[Cached]** tag appears next to the Source column whenever a result was served from cache.

Actions — Search box, IPO filter (for multi-IPO runs), “Download Excel” button.

Dashboard

Shown after a run completes:

Total Clients	Allotted	Not Allotted	Errors	Served from Cache
1,200	320	860	20	180

When multiple IPOs were selected, a compact per-IPO breakdown table is added.

History

- List of past runs: date, IPO(s), file name (or client identifier), result summary.
- Selecting a past run reopens its results/report without re-checking any client (PRD FR-017).

Notifications

- On completion: a toast/banner such as “Completed — 1,000 Clients × 2 IPOs, 20 Errors, 180 from Cache — Download Report”.

Component Inventory

Component	Used In	Notes
IpoMultiSelect	Single Client, Bulk Upload	Multi-select dropdown; shows “Auto-detected” badge; selected IPOs render as removable chips

Component	Used In	Notes
IdentifierInput	Single Client	Inline PAN/Client Code format validation
FileDropzone	Bulk Upload	Drag-and-drop + file picker, type/size enforcement
ColumnMappingSummary	Bulk Upload	Shows detected → mapped column pairs
ValidationSummary	Bulk Upload	Valid/invalid row counts with per-row reasons, plus selected-IPO count
ProgressBar	Progress Screen	Live completed/total + current client/IPO + running cache-hit count
CaptchaPrompt	Progress Screen	Inline CAPTCHA image + submit input, shown only in Assisted Mode
ResultsTable	Results Screen	Sortable/searchable, status badges, IPO filter, Source column with Cache tag
StatusBadge	Results Screen	Color-coded status categories
SourceTag	Results Screen, Dashboard	Distinguishes a live registrar source from a “Cached” result
DashboardStats	Dashboard	Summary counts, including cache-served count and per-IPO breakdown
HistoryList	History	Past runs, reopen action, multi-IPO indicator
NotificationToast	Global	Run-completion summaries

Interaction Flow Summary

Home → [Single Client] → Enter PAN/Code + Select IPO(s) → Check → Result(s) (per IPO)

Home → [Bulk Upload] → Upload file → Select IPO(s) → Column mapping + validation
→ Confirm → Progress screen (aggregate + per-IPO) → Results + Dashboard
→ Filter by IPO / Search / Download Excel / (later) revisit via History

Accessibility and Usability Notes

- Status categories are distinguished by both color and icon/text, not color alone. The same applies to the Cached tag.
- Validation errors are shown inline, next to the offending field or row.
- The progress screen must remain responsive throughout a bulk run, even when a run spans several IPOs.
- When multiple IPOs are selected, the UI must make the scope of the run unambiguous at every stage.

Future Frontend Enhancements

- Authenticated views reflecting role-based permissions (admin vs. operator) once login is introduced.
- In-app notification center once email/WhatsApp notifications are introduced on the back-end.
- User-configurable cache TTL visibility once exposed by the backend.